ModelIt!

Student Activity Pack

# Is Social Media Rewiring Your Brain?

Modeling Dopamine Pathways, Attention Regulation, and Digital Addiction Mechanisms

Grade 12L1 | NGSS HS-LS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How many times do you estimate you check your phone each day? Do you think the actual number is higher or lower than your estimate?**

2. **Have you noticed any changes in your ability to focus on reading or homework over the past few years? What do you think caused those changes?**

3. **Why do you think scrolling through TikTok or Instagram feels so satisfying in the moment but often leaves you feeling worse afterward?**

4. **Do you think social media companies have a responsibility to make their products less addictive? Why or why not?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Social Media Exposure | Notification Frequency | Dopamine Sensitivity | Attention Capacity | Cognitive Well-Being

| EXTERNAL | INTERNAL |
|---|---|
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Doom Scrolling

Social Media Exposure: 4+ hours passive scrolling | Notification Frequency: Maximum | Duration: 30 days

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Mindful Usage

Social Media Exposure: 1 hour active/creative | Notification Frequency: Off | Duration: 30 days

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Recovery Period

Social Media Exposure: Zero | Notification Frequency: Zero | Duration: 14 days after heavy use

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Social Comparison Intensity** — The degree to which social media use triggers upward social comparison — comparing yourself to curated highlight reels of others' lives, which activates stress pathways and reduces self-esteem independently of the dopamine and attention effects

**Sleep Disruption** — The impact of evening screen use on melatonin production and sleep architecture — blue light suppresses melatonin, social media content increases arousal, and poor sleep independently impairs dopamine system recovery and attention restoration

**FOMO Activation** — Fear of missing out — a psychological driver that keeps users checking social media even when they want to stop, operating through anxiety pathways that are separate from but synergistic with the dopamine reward system

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**You check your phone 96 times a day — once every 10 minutes. Instagram, TikTok, and Snapchat were designed by neuroscientists to be as addictive as possible. Is social media physically changing the structure and function of your brain, and if so, can you get it back?** Use evidence from your simulation to support your answers.

**1 How does your model explain why TikTok and Instagram feel more compelling than reading a book, even when you know the book is more valuable?**

**2 What does dopamine receptor downregulation mean for your ability to enjoy non-digital activities like conversation, nature, or hobbies?**

**3 Your model shows that notification frequency is a powerful driver of compulsive checking. What would happen if you turned off ALL notifications for a week?**

**4 Should social media platforms be required to disclose the psychological manipulation techniques they use, similar to nutrition labels on food?**

**5 If social media is rewiring teenage brains during critical development periods, is this a personal responsibility issue or a public health issue?**

# 8 STEM Challenge: Design a Digital Wellness App

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## YOUR ENGINEERING MISSION

Design a smartphone application that helps teens monitor and manage their social media use by providing real-time feedback on neurological effects, using the same behavioral neuroscience principles that make social media addictive — but in reverse, to build healthier digital habits.

## THE SCENARIO

A tech company wants to create a counter-product: an app that uses behavioral neuroscience to help people break unhealthy social media habits. The app must be based on the same dopamine and reinforcement science that makes platforms addictive, but designed to promote digital wellness. The challenge: the app itself must avoid becoming another addictive screen experience.

### Engineering Design Thinking

1. WHAT SPECIFIC BEHAVIORAL NEUROSCIENCE PRINCIPLES DOES YOUR APP USE TO COUNTERACT ADDICTIVE SOCIAL MEDIA DESIGN?

2. HOW DOES YOUR APP PROVIDE FEEDBACK WITHOUT BECOMING ANOTHER SOURCE OF COMPULSIVE CHECKING AND DOPAMINE-SEEKING?

3. WHAT MODEL EVIDENCE SUPPORTS YOUR CLAIM THAT YOUR APP'S INTERVENTION STRATEGY WILL IMPROVE ATTENTION CAPACITY AND DOPAMINE SENSITIVITY?

4. HOW WOULD YOU MEASURE WHETHER YOUR APP ACTUALLY REDUCES PROBLEMATIC SOCIAL MEDIA USE AND IMPROVES COGNITIVE WELL-BEING?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Behavioral neuroscientists study how digital technology affects brain function and development, earning $70,000-$140,000/year in research positions. UX ethics designers ensure technology products don't exploit psychological vulnerabilities, earning $90,000-$160,000/year. Digital wellness consultants help organizations develop healthy technology policies, earning $65,000-$120,000/year.